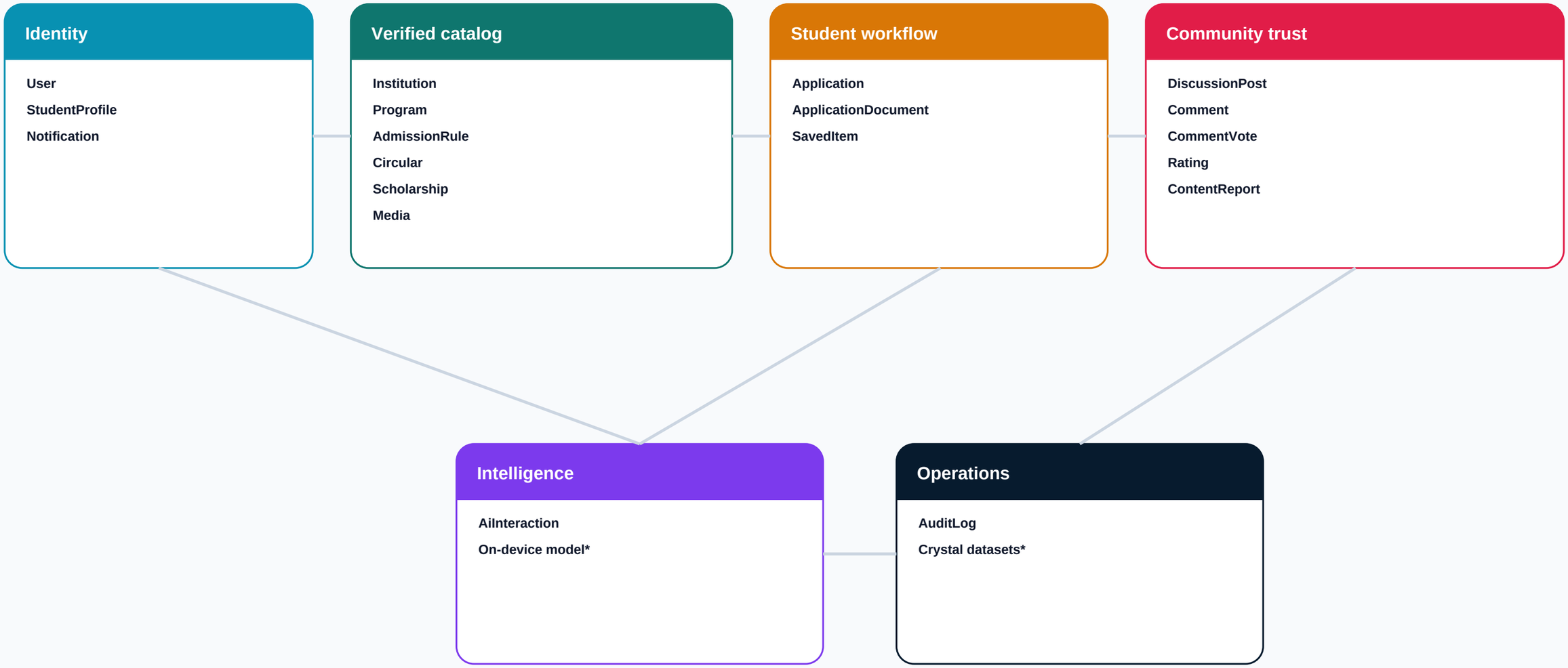
# EduPath BD - Entity Relationship Diagram

System domain map

## Relational architecture overview

Arrows indicate the main data flow; full cardinalities and foreign keys are listed on page 4.




### Design guarantees

- One account per verified identity
- Official-source fields separated from user opinion
- One vote/rating per account
- Moderation and AI audit trail
- Cross-platform client; server-owned security

** Non-table runtime/reporting components are shown for system context and are not database entities.*

# EduPath BD - Identity and Catalog Entities

Entity keys and important attributes

### User

PK id
UK username
UK email
role
emailVerifiedAt
phoneVerifiedAt

### StudentProfile

PK id
FK/UK userId
currentStage
academicGroup
sscGpa / hscGpa
interests[]
careerGoals[]

### Institution

PK id
UK slug
type
district
verificationStatus
sourceUrl
latitude / longitude

### Program

PK id
FK institutionId
UK institutionId + slug
level
tuitionMin / tuitionMax
careerOutcomes[]
verificationStatus

### AdmissionRule

PK id
FK programId
UK programId + session
allowedGroups[]
minimum GPA fields
subject rules (JSON)
sourceUrl

### Circular

PK id
FK institutionId
FK programId (nullable)
academicSession
applicationEndsAt
applicationFee
sourceUrl
verificationStatus

### Scholarship

PK id
FK institutionId (nullable)
FK programId (nullable)
provider
eligibility (JSON)
deadline
verificationStatus

### Application

PK id
FK userId
FK programId
UK userId + programId
status
referenceNo
appliedAt

### ApplicationDocument

PK id
FK applicationId
title
fileUrl / storagePath
isRequired
isCompleted
uploadedAt

### DiscussionPost

PK id
FK authorId
FK institutionId (nullable)
FK programId (nullable)
FK circularId (nullable)
title / body
isLocked

Legend: PK = primary key, FK = foreign key, UK = unique constraint. Nullable relationships are explicitly labelled.

# EduPath BD - Community, Workflow and AI Entities

Entity keys and important attributes

### Comment

PK id
FK postId
FK authorId
FK parentId (self, nullable)
body
isVerified
isHidden

### CommentVote

PK id
FK commentId
FK userId
UK commentId + userId
type: LIKE / DISLIKE

### Rating

PK id
FK userId
FK institutionId (nullable)
FK programId (nullable)
UK user + target
score
review

### Media

PK id
FK institutionId (nullable)
FK programId (nullable)
type
title
url

### SavedItem

PK id
FK userId
UK user + itemType + itemId
itemType
itemId
createdAt

### Notification

PK id
FK userId
title / body
type
data (JSON)
scheduledAt
readAt

### AiInteraction

PK id
FK userId
feature
prompt / response
model
sourceIds[]
safetyNotice

### ContentReport

PK id
FK reporterId
UK reporter + target
targetType / targetId
reason
status
resolution
reviewedAt

### AuditLog

PK id
FK adminUserId
action
entityType / entityId
before / after (JSON)
createdAt

Legend: PK = primary key, FK = foreign key, UK = unique constraint. Nullable relationships are explicitly labelled.

# EduPath BD - Relationship Matrix

Cardinalities, keys and delete behavior

| Parent | Cardinality | Child | Foreign key | On delete |
|---|---|---|---|---|
| User | 1 - 0..1 | StudentProfile | StudentProfile.userId | CASCADE |
| Institution | 1 - many | Program | Program.institutionId | CASCADE |
| Program | 1 - many | AdmissionRule | AdmissionRule.programId | CASCADE |
| Institution | 1 - many | Circular | Circular.institutionId | CASCADE |
| Program | 1 - many | Circular | Circular.programId | SET NULL |
| Institution | 1 - many | Scholarship | Scholarship.institutionId | SET NULL |
| Program | 1 - many | Scholarship | Scholarship.programId | SET NULL |
| User | 1 - many | Application | Application.userId | CASCADE |
| Program | 1 - many | Application | Application.programId | CASCADE |
| Application | 1 - many | ApplicationDocument | ApplicationDocument.applicationId | CASCADE |
| User | 1 - many | DiscussionPost | DiscussionPost.authorId | CASCADE |
| Institution | 1 - many | DiscussionPost | DiscussionPost.institutionId | CASCADE |
| Program | 1 - many | DiscussionPost | DiscussionPost.programId | CASCADE |
| Circular | 1 - many | DiscussionPost | DiscussionPost.circularId | CASCADE |
| DiscussionPost | 1 - many | Comment | Comment.postId | CASCADE |
| User | 1 - many | Comment | Comment.authorId | CASCADE |
| Comment | 1 - many | Comment | Comment.parentId | CASCADE |
| Comment | 1 - many | CommentVote | CommentVote.commentId | CASCADE |
| User | 1 - many | CommentVote | CommentVote.userId | CASCADE |
| User | 1 - many | Rating | Rating.userId | CASCADE |
| Institution | 1 - many | Rating | Rating.institutionId | CASCADE |
| Program | 1 - many | Rating | Rating.programId | CASCADE |
| Institution | 1 - many | Media | Media.institutionId | CASCADE |
| Program | 1 - many | Media | Media.programId | CASCADE |
| User | 1 - many | SavedItem | SavedItem.userId | CASCADE |
| User | 1 - many | Notification | Notification.userId | CASCADE |
| User | 1 - many | AiInteraction | AiInteraction.userId | CASCADE |
| User | 1 - many | ContentReport | ContentReport.reporterId | CASCADE |
| User | 1 - many | AuditLog | AuditLog.adminUserId | RESTRICT |

## Polymorphic references

- SavedItem.itemId
- ContentReport.targetId
- AuditLog.entityId
- AiInteraction.sourceIds[]
- Notification.data (JSON)

## Integrity rules

- Unique usernames and emails
- One application per program
- One vote per comment
- One rating per target
- One report per user/target
- Cascade ownership cleanup

## Source of truth

Schema:

apps/api/prisma/schema.prisma

Generated:

Phase 5 checkpoint

Database: PostgreSQL